

Amazon Q Developer

Summary: -

I learned and used Amazon Q Developer, an AI coding assistant that helps developers write, improve, test, and fix code. I connected it with VS Code, signed in using AWS Builder ID, and used it to generate code, explain programming concepts, and support different stages of software development like planning, coding, testing, deployment, and maintenance.

Amazon Q Developer: -

- Amazon Q Developer is an AI-powered assistant that helps developers build, analyze, and manage applications on AWS by generating code, answering questions, and improving software quality.
- It can Assist in coding and testing.
- It supports all the major programming language like java, python, Go etc.
- It helps in diagnosing errors
- It can also perform security scanning and fixes.
- Where to Access Amazon Q – Developer from: -
 - AWS Management Console
 - AWS CLI
 - IDE such as VS code

How Amazon Q can helps in Software Development Lifecycle: -

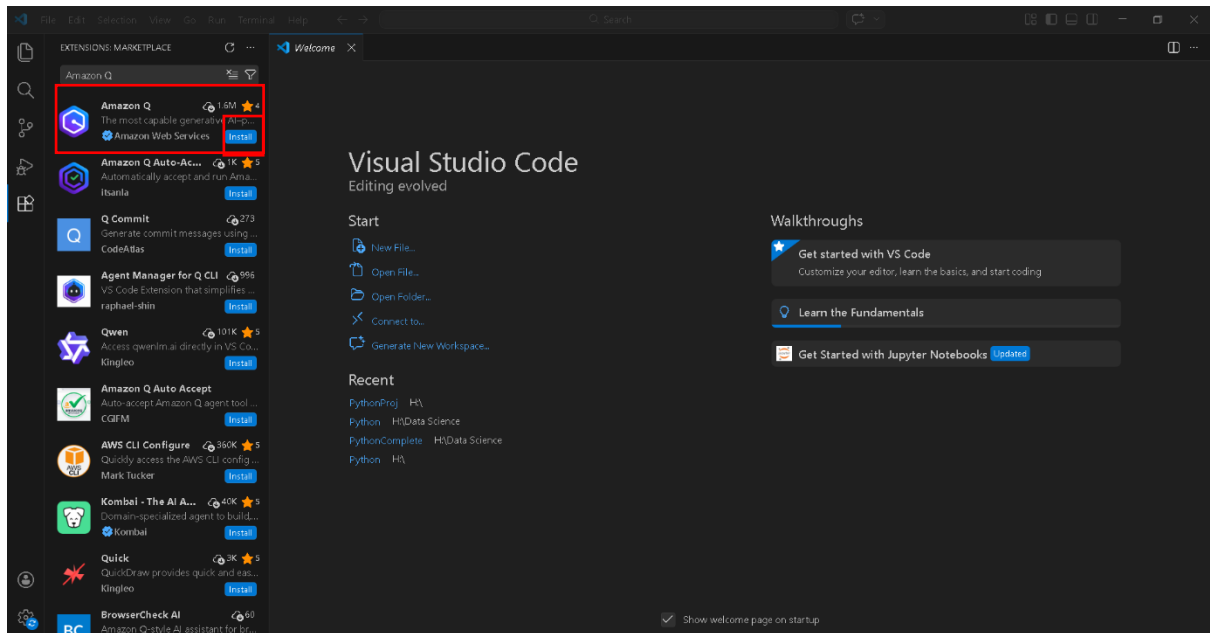
Planning: -

- Business Requirement Document(Epic) : -
 - Ex: - A store inventory manager, I should be able to check the inventory details for items in the store and update them when new items are received at the store.
 - Features and User Stories: -
 - Based on Epic we create our Features and User Stories.
-
- For this phase we can use Amazon Q Business
 - In Amazon Q Business Jira Plugin and you can write the requirements in the Natural language in the chat interface.
 - And then it creates Features and User stories automatically in the JIRA.

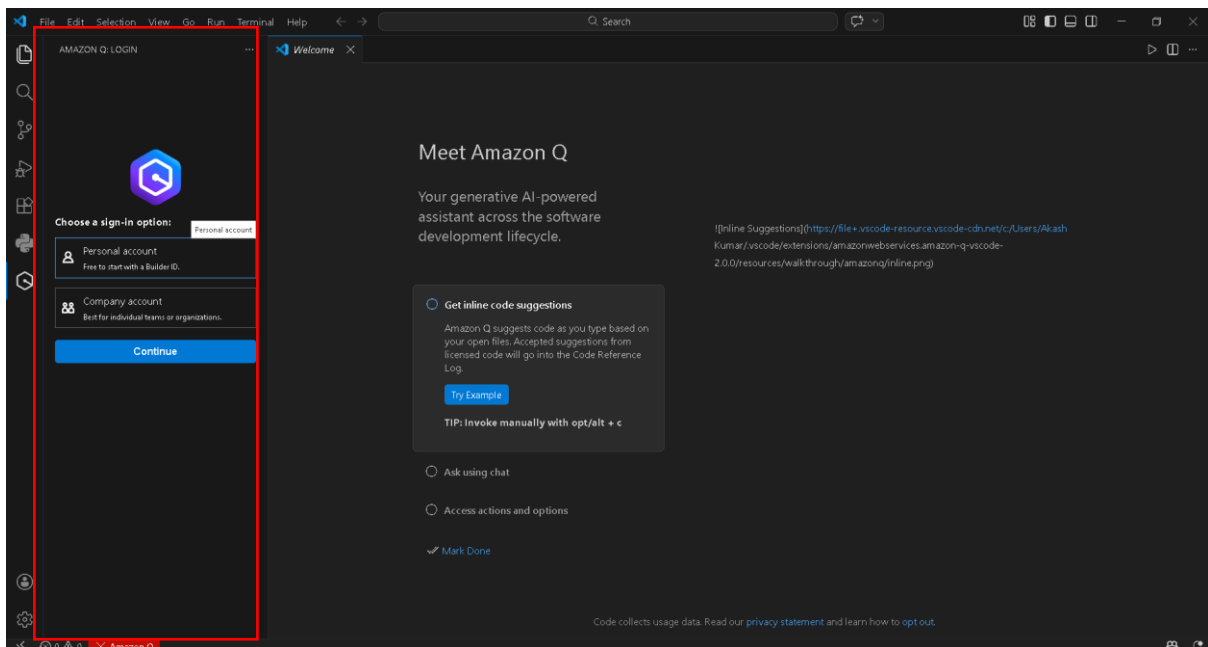
Design: -

In the Design Phase Amazon Q Developer can help you generate Architecture for the use case.

So now Let's see how: -



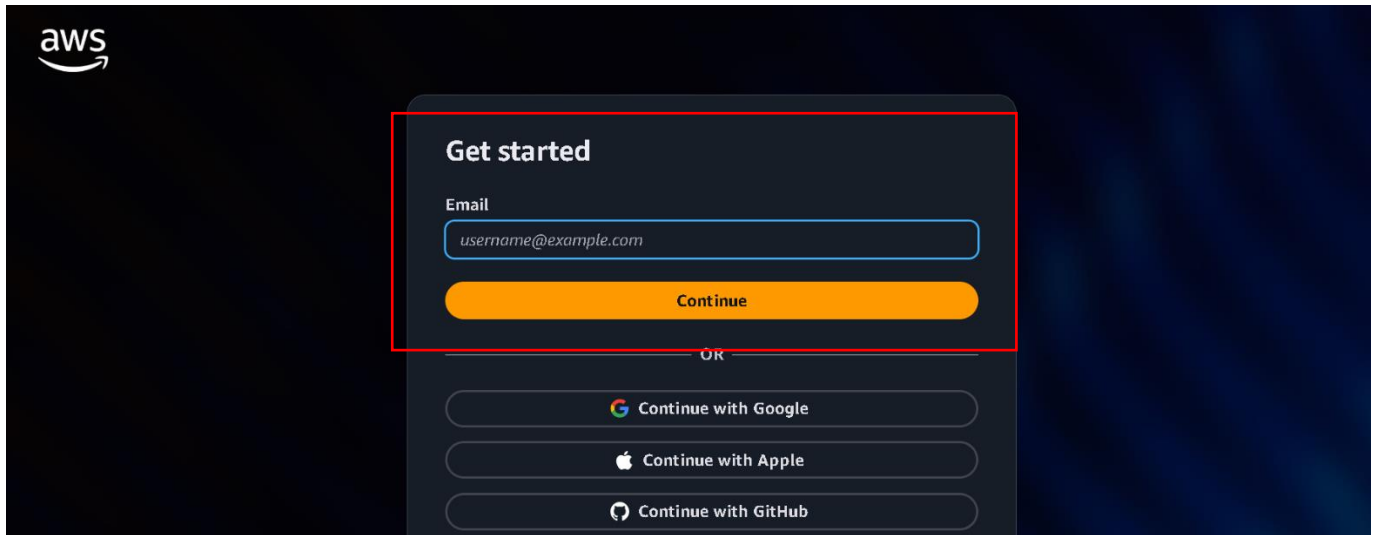
- The first thing you need to download VS Code.
- Open it and click on Extension icon (left side Fifth icon)
- And Search Amazon Q
- Click on it and install it



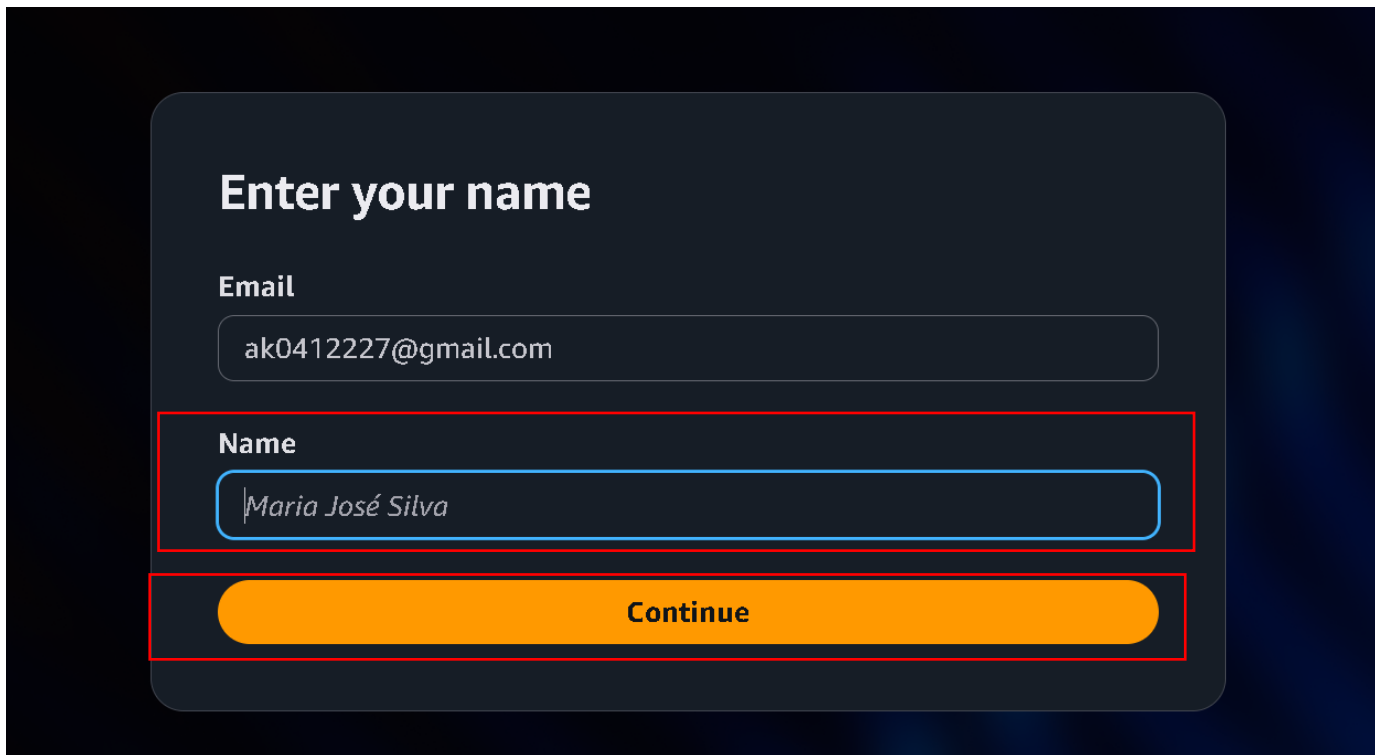
- Click on it and install it
- Click on sign in with personal account.
- But for sign in you need a Builder id

So let's create a Builder id: -

Open any Browser and search AWS Builder ID.



- Search on browser create AWS Builder ID.
- Click on the first Link.
- Then Enter the Email.
- You can choose continue with google, Apple etc.



- Now Enter Name.
- And click continue.

Verify your email

We just sent a verification code to your email. It may take up to 5 minutes for the code to arrive.

Email

fecaja5601@ryzid.com [Change](#)

Verification code

[Resend code 27s](#)

Code expires in 29:27 minutes

[Continue](#)

- Now Enter Verification code.
- And click continue.

Create your password

Email

fecaja5601@ryzid.com

Password

Enter password

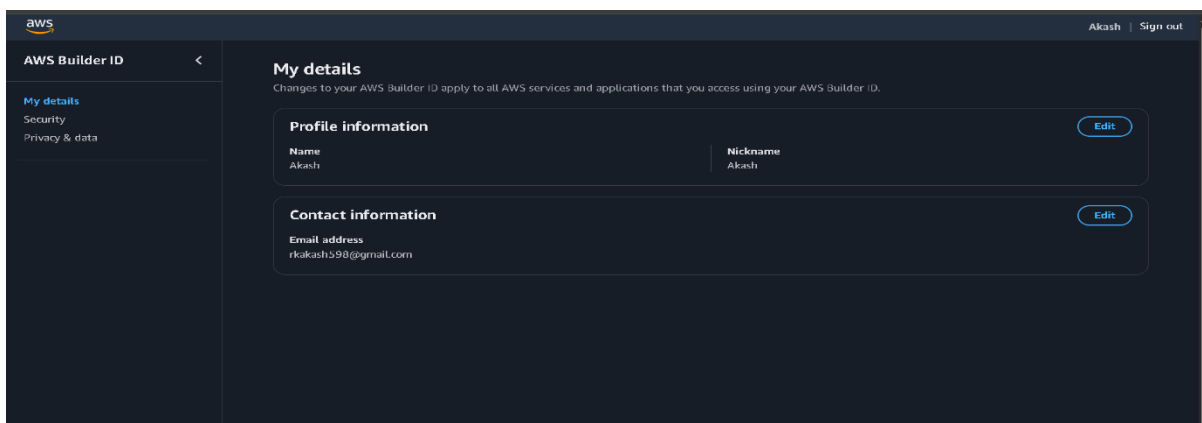
Confirm password

Re-enter password

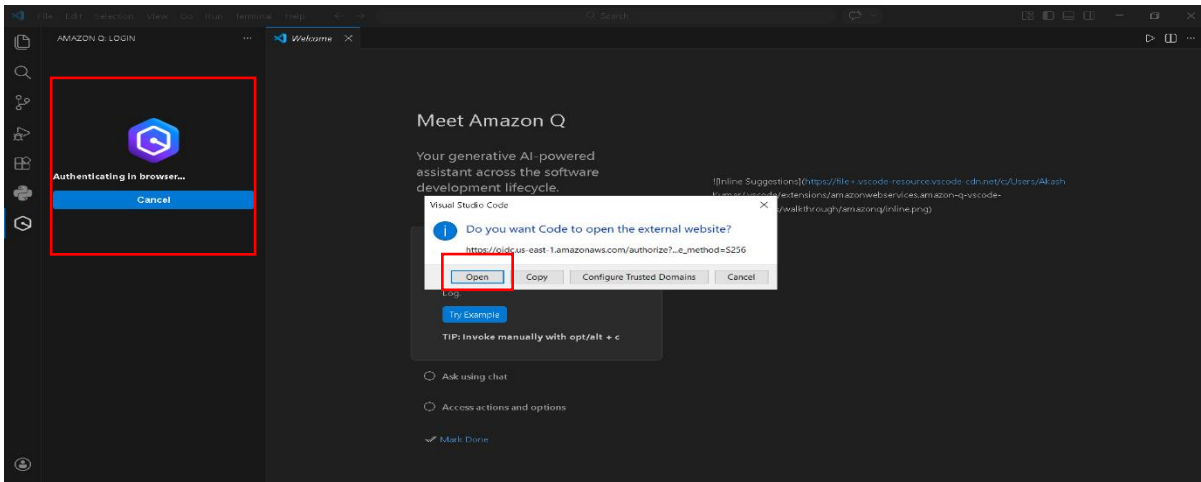
☐ Show password

Continue

- Now Enter Password and confirm password.
- And click Continue.
- Now your account is created.



- Now your Builder id is Ready
- Now Back to your Vs Code and go in Amazon Q
- Choose Sign in personal account
- Click continue.



- Now click on Open
- Redirected to your browser



Allow Amazon Q Developer for command line to access your data?

By choosing **Allow** access, you agree to allow **Amazon Q Developer for command line** to access the following:

 **Kiro**
[Show details](#)

[Deny access](#) [Allow access](#)

- Now click Allow Access.

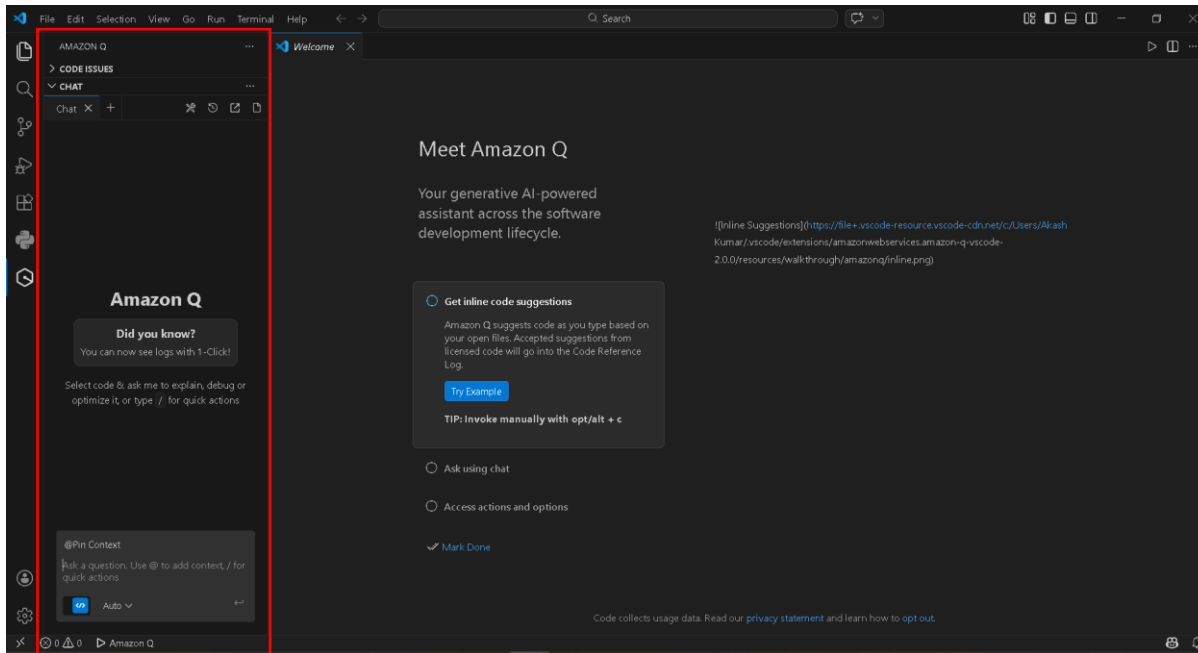


✓ **Request approved**

Amazon Q Developer for command line can now access your data in Kiro.

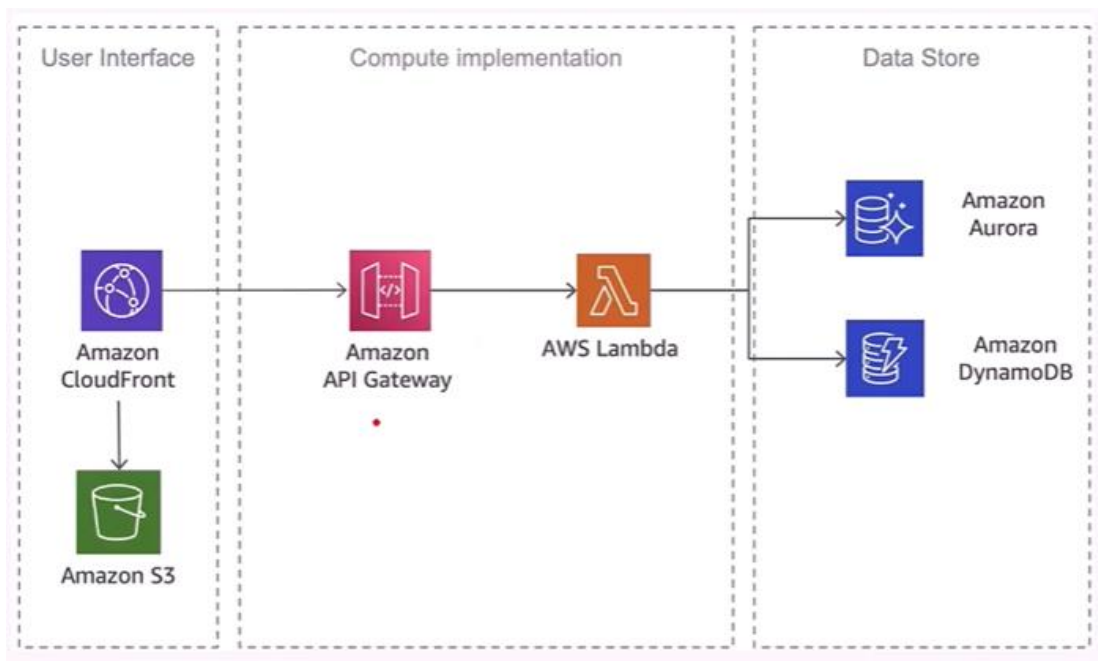
You can close this window.

- Now Request approved.



- Now our Amazon Q is ready to ask any question it's a LLM

Now Understand about AWS Micro-Services Architecture: -

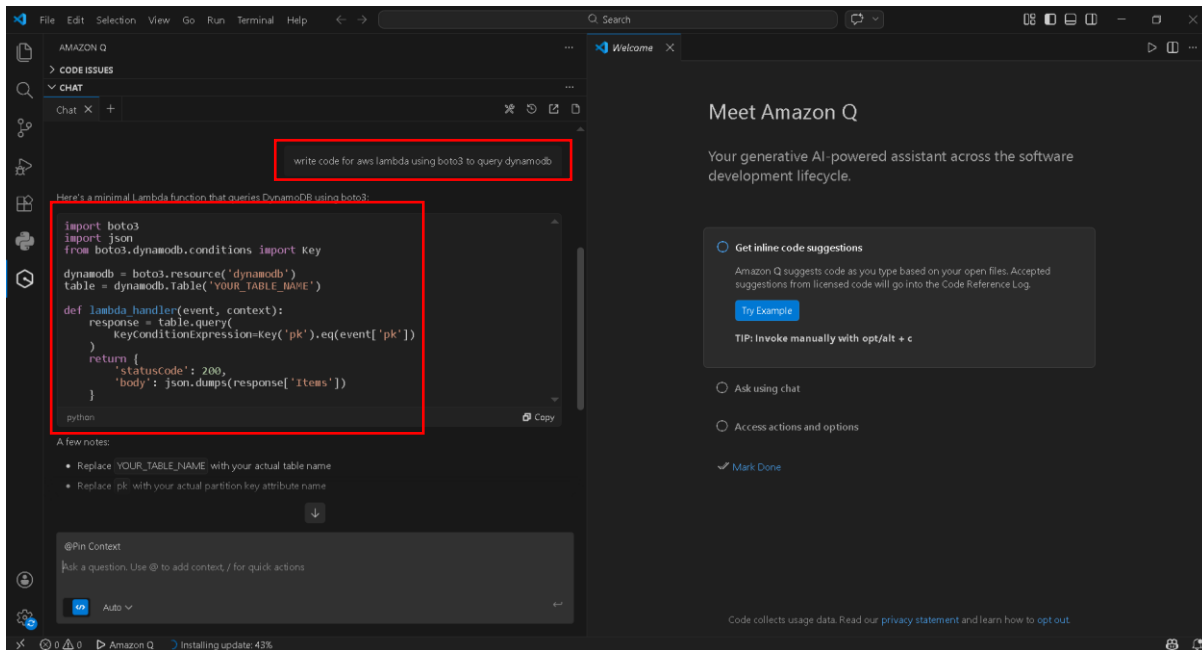


- Now this is our Flow: -
 - We will Send the request to Amazon API gateway
 - Which invokes the Amazon Lambda Function
 - And then Amazon Lambda makes a query to the Amazon Dynamo DB
- Implementation: -
 - Coding
 - Debugging
 - Improve code

➤ Unit Testing

- **Example : -Now let's generate the code for this Architecture: -**

- Now let's we start coding
- In which we'll have write the Lambda function which will query the Amazon Dynamo DB
- And we are going to use python SDK Boto3 for doing this.
- let's Ask Amazon Q Developer to Generate the code



- Now you see it write code for us.
- Also explain and fix your already written code.

- **Testing: -**

- Security Testing
- Functional Testing
- Defects

- **Deploy: -**

- Infra setup
- Code Deployment

- **Maintain: -**

- Defect
- RCA

➤ Transform



NOTE: - This Note to give you a overview of how Amazon Q Developer can help across software development.